

aviation where a blend of capital investments and tech innovation is required to meet short-term and long-term goals. “In India, we have developed a robust supply chain for aero engine components in our Pune factory and with our 13 other commercial engine parts manufacturers, including Tata Advanced Systems, who are integral to our global supply chain,” he says.

Experience and a sound understanding of the space shape how large MNCs use tech in the domestic markets. Mathur of Siemens says its digitalisation and automation technologies have enabled local manufacturers to compete on a global scale. Citing the example of an automotive customer, he points out how a four-decade old plant was able to boost productivity by 20% by adopting digital technologies. “A large F&B company increased its throughput by 1.5% and had energy savings of 4%,” he adds.

WHAT NOW?

There are other issues that need to be sorted on the path to 2047. Avalon’s Sengupta speaks of India having struggled with policy clarity. “Industry lobbies have often succeeded in getting duty structures inverted, leading to Bangladesh stealing a march in garments since domestic textiles needed protection. That policy has affected solar, gas, electronics, paper or furniture and we will need a clearer framework.”

Then, there is the issue of skills. GE Aerospace’s Nagar points out that India has diploma and graduate engineers with a general knowledge of manufacturing processes. “Aero engine manufacturing requires specific metal joining skills that are not currently taught,” he says. “Manufacturing training courses in this area could help in dealing with the issue of talent shortage.”



“The tipping point will emerge when global supply chains also relocate to India to feed into Indian manufacturing”

SUNIL MATHUR
MD & CEO, SIEMENS LTD



“Gen AI tells me the propensity of a risk/accident and that is done in seconds. Likewise, if I am interviewing someone, it throws up a lot of relevant information about that person”

JAYANTA BANERJEE
GROUP CIO, TATA STEEL

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Sengupta maintains manufacturing automation will continue with AI increasingly integrated in the processes. “That integration would be in the areas of quality, including online checking of key output parameters and tweaking of input variables. As AI’s capability increases, it will result in increased flexibility in manufacturing as well,” he says, adding that the incorporation of AI-enabled tools will also lead to a reduction in the carbon footprint. He cites examples of how large companies such as Hindalco Industries have recycled aluminium, Vedanta’s initiatives in using automation

and advanced control systems in smelting processes, and JSW Steel adopting green technologies resulting in electric arc furnaces and utilising waste gases produced for auxiliary purposes.

India has come a long way in the manufacturing space since Independence, but it has also missed many opportunities on the way. Experts say now is the time to correct that and make the best use of the present for a brighter future. It is too big an opportunity for the country to miss. **BT**

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